EXP NO: 4
DATE: 27/8/26

Setting Up a Local Area Network.

Aim

To design, configure and verify a 4-PC Local Area Network (LAN) using a central switch, static IP addressing, ping connectivity tests and HTTP resource sharing.

Equipment Required:

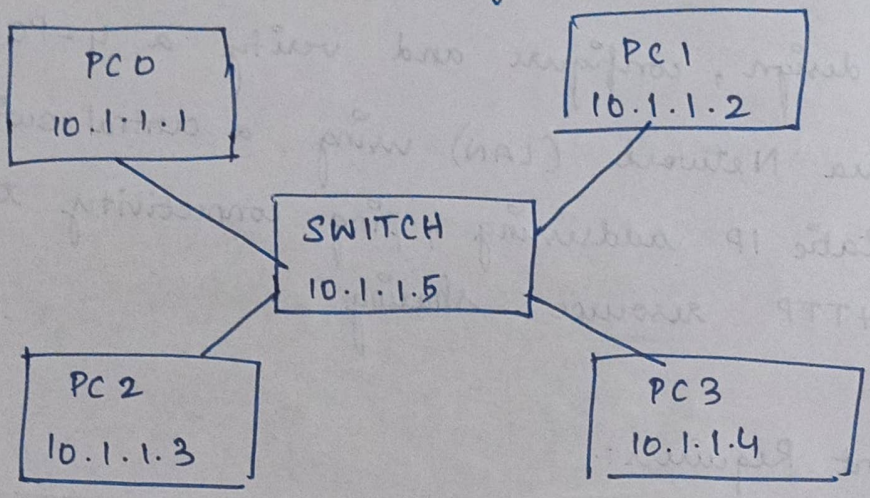
Equipment	Quantity	Specification
Personal Computer	4	Ethernet (NIC)
Network Switch	1	8/16/24 port
Ethernet Cables	4	straight-through

Theory:

A Local Area Network (LAN) connects devices within a limited area to share resources. This experiment uses a Star Topology where host connect to a central Layer 2 switch. The switch forwards data frames using MAC addresses. Devices communicate by sharing an IP network range defined by their subnet mask

Network Diagram :

Star Topology LAN



IP Configuration table :

Device	IP Address	Subnet Mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/
PC1	10.1.1.2	255.0.0.0	Switch Port Fa0/
Pc 2	10.1.1.3	255.0.0.0	Switch Port Fa0
PC 3	10.1.1.4	255.0.0.0	Switch Port Fa0/
Switch	10.1.1.5	255.0.0.0	VLAN

Implementation Steps :

1. Place Devices :

Drag 4 PC, 1 Server and 1 Switch (2960) to the workspace.

2. Cable Connections :

Connect PCs (Fa0/1 to Fa0/4) and server (Fa 0/5) to switch using

copper straight through cables.

3. Assign Host IP's:.

Set static IP addresses and subnet mask (255.0.0.0) on all PC's and the Server.

4. Configure Switch Management:.

Open Switch CLI and run

```
enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit.
```

5. Verify Connectivity :.

Run ping 10.1.1.2 from PC0 Command Prompt to verify network link.

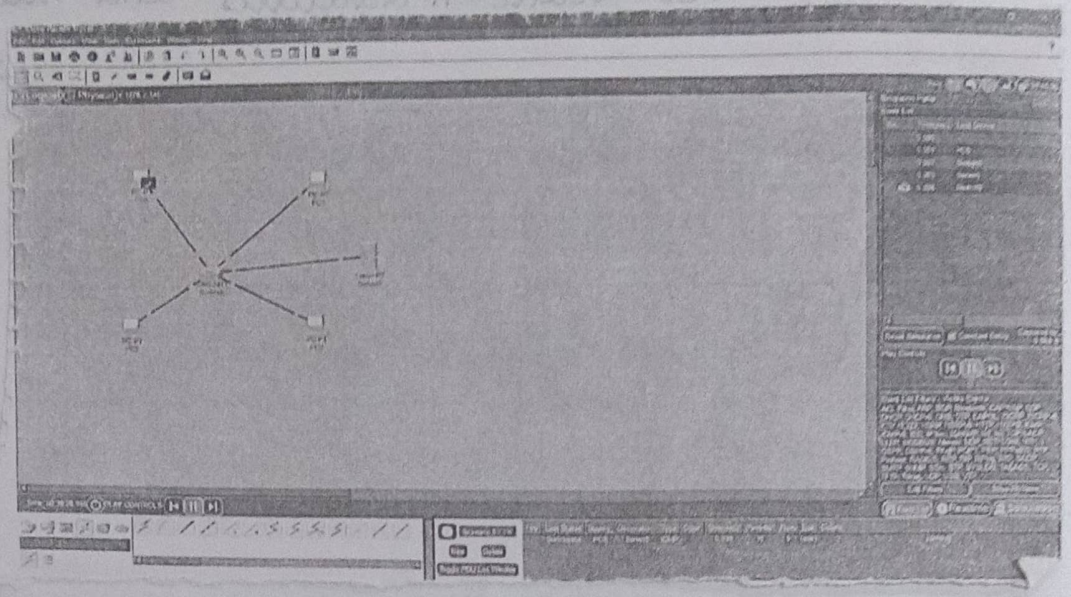
6. Demonstrate Resource sharing:.

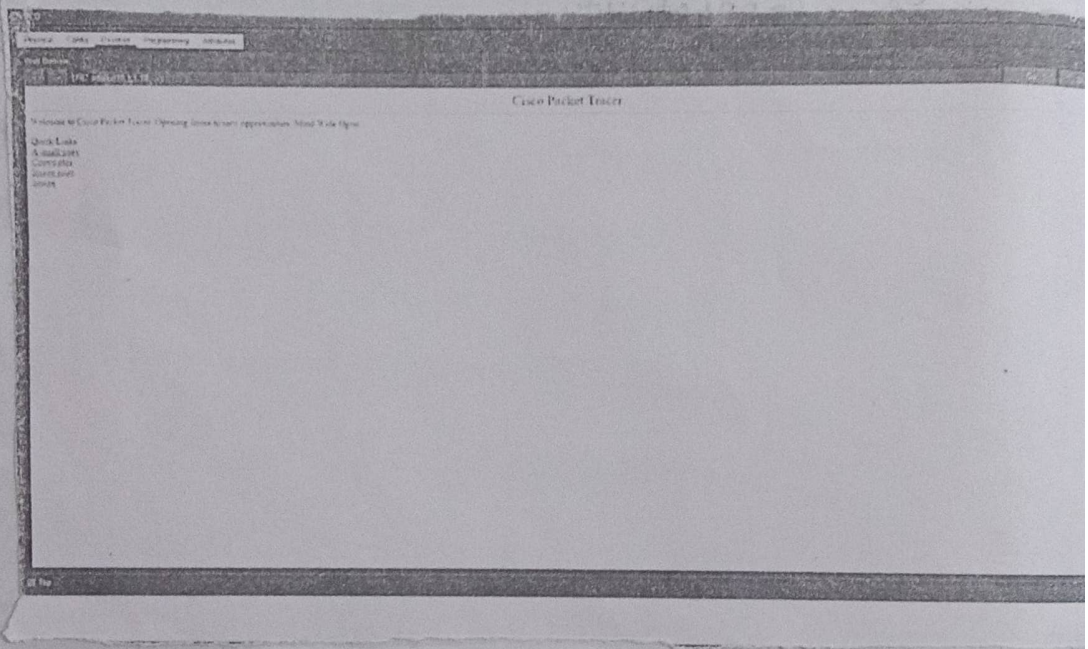
Enable HTTP on Server (10.1.1.10). Open client PC Web browser and navigate to [hHp :// 10.1.1.10].

7. Observe Packet flow:.

Switch to Simulation mode to trace ARP and ICMP/HTTP frame forwarding ~~accross~~ across ports.

Output :-





Result :-

The LAN lab experiment was successfully built, configured, and verified for IP connectivity and client - server resource sharing.